Aarhus University

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Introduction to Algebra 2

Handin 1

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Exercise 13.5

Show Theorem 13.3 [1].

Show first that $K[X]$ has at least one monic irreducible polynomial.

First, a quick definition of what we are working with:

1. K is a field, see Definition 12.4 [1].
2. $K[X]$ is a polynomial ring, see Chapter 13.1 [1].
3. Monic is when the leading coefficient of a polynomial is 1, see Chapter 13.1 [1].
4. Irreducible is when every factorization of a non-constant polynomial in $K[X]$ only has trivial factorizations (where one factor is a unit), see Section 13.2 [1].

Let's start by showing that a monic polynomial exists in $K[X]$.
Per the definition of a ring 12.1(f) [1], there exists a neutral element $1_K \in K$.
Now, let there be a polynomial of the first degree, X^1

$$X^1 \cdot 1 = 1 \cdot X^1 = X^1$$

Here, it holds that the leading coefficient is 1, so the polynomial is monic!

Now we must show that the monic polynomial X^1 is also irreducible.
This is done by showing that all factorizations

$$X^1 = f(X) \cdot g(X) \tag{1}$$

are trivial. A factorization is called trivial if one of the factors is a unit, definition 12.8 [1].
According to Theorem 13.1(a) [1], it holds that the degrees of $f(X)$ and $g(X)$ must sum to the degree of X^1:

$$\deg f(X) + \deg g(X) = 1 \qquad \text{for } \deg f(X), \deg g(X) \geq 0$$

This means at least one of them must have a degree of 0 and be a constant. It holds from Theorem 13.1(c) [1], a polynomial in $K[X]$ is a unit if and only if it is a constant and a unit in K.
K is a field where all non-zero elements are units (Definition 12.4) [1], so one of the factors must be a unit in $K[X]$.
This makes all factorizations in (1) trivial, making X^1 irreducible.
This shows that there is AT LEAST ONE monic irreducible polynomial in $K[X]$.

Show that the assumption that a finite number of irreducible monic polynomials exists leads to a contradiction

This can be shown using the Euclids proof from 5.1 [1].
Assume there is a finite number of irreducible monic polynomials, $P_1(X), P_2(X) \dots P_n(X)$.
Construct a new polynomial $F(X)$

$$F(X) = P_1(X) \cdot P_2(X) \cdots P_n(X) + 1$$

This new polynomial is monic according to Theorem 13.1(a) [1], the leading coefficient of $F(X)$ is the product of the coefficient of all its polynomial factors, which we know to be 1, since they're all monic, hence $F(X)$ is monic
Now from Theorem 13.2 [1], and because $\deg F(X) > 0$, $F(X)$ then either $F(X)$ is irreducible or one of its factors must be irreducible, in both cases a monic irreducible factor exists, let's call it $q(X)$.

This new factor $q(X)$ must be in the the list $P_1(X), P_2(X) \dots P_n(X)$ since it's assumed to contain all monic irreducible polynomials, so $q(X) = P_i(X)$ for some $i \in \{1, 2, \dots, n\}$.
This leads to the following

$$\begin{aligned} & P_i(X) \mid F(X) \wedge P_i(X) \mid P_1(X) \cdot P_2(X) \cdots P_n(X) \\ \implies & P_i(X) \mid F(X) - (P_1(X) \cdot P_2(X) \cdots P_n(X)) \\ \implies & P_i(X) \mid 1 \end{aligned}$$

This leads to a contradiction since the polynomials that can divide 1 is the constant polynomials, but since $P_i(X)$ is irreducible it can't be constant, which means the assumption of there being a finite number of monic irreducible polynomials must be false and there must be an infinite amount of monic irreducible polynomials.

References

[1] Johan P. Hansen. *Tal og polynomier. Begreber, metoder, resultater, kodning og kryptografi*. Aarhus Universitetsforlag, 2018.